HAND & UPPER LIMB · NEET-SS / FET / INI-SS

Flexor and extensor tendons

1. Flexor tendon anatomy

- **FDS** splits at the proximal phalanx (**Camper's chiasma**) and inserts on the **middle phalanx**; **FDP** passes through the split to the **distal phalanx base**. **FDP of index acts independently** (separate muscle belly); middle/ring/little FDP share a common belly (**quadriga effect**).
- **FDS to the little finger is absent or weak in about 15-20%**.
- **Flexor sheath pulleys: five annular (A1-A5)** and **three cruciform (C1-C3)**; **A1, A3, A5 over joints (MCPJ, PIPJ, DIPJ)**; **A2 (proximal phalanx) and A4 (middle phalanx)** are **the most important biomechanically** (prevent **bowstringing**). **Thumb:** A1, **oblique pulley (most important)**, A2.
- **Nutrition: synovial diffusion** (major within the sheath) and **vincula** (**vinculum longum and breve**, carry blood via transverse digital arteries); **watershed avascular zones** on the volar aspect of the tendons.

2. Flexor tendon zones (Verdan/Kleinert)

Zone	Extent	Notes
I	Distal to FDS insertion (FDP only)	**Jersey finger** (FDP avulsion)
II	**A1 pulley (distal palmar crease) to FDS insertion**	**"No man's land"** (Bunnell): **both FDS and FDP within the tight fibro-osseous sheath**; adhesions; historically poor results
III	Distal edge of carpal tunnel to A1 pulley	**Lumbrical origins** (palm); better prognosis

Zone	Extent	Notes
IV	**Carpal tunnel**	Median nerve injury associated
V	Proximal to carpal tunnel (forearm)	Multiple tendons, nerves and arteries ("**spaghetti wrist**")
Thumb T I-T V	T I distal to IPJ; **T II A1 to IPJ**; T III thenar eminence; T IV carpal tunnel; T V forearm	FPL

3. Flexor tendon repair

- **Timing: primary repair (within 24 hours)** or **delayed primary (within about 10-14 days)**; beyond **3-4 weeks**: tendon retraction and myostatic contracture: consider **staged reconstruction**.
- **Principles:** atraumatic handling, **multi-strand core suture** (**strength is proportional to the number of strands crossing the repair site**: **4-strand or 6-strand core** (e.g. **Adelaide, cruciate, modified Kessler/Strickland**) allows **early active motion**), **core purchase about 7-10 mm from the cut end**, **dorsally placed core suture** is stronger, **epitendinous running suture** (adds about 10-50% strength and reduces gap/gliding resistance), **locking loops** add strength.
- **Gap formation over 3 mm** at the repair: increased **rupture and adhesions**.
- **Repair strength:** decreases between **days 5 and 21** (softening phase) and recovers after that.
- **Pulleys:** preserve **A2 and A4**; **venting (partial release) of A2/A4 (up to about two-thirds of A2 or the whole A4 if needed)** allows passage of bulky repairs without significant bowstringing.
- **FDS in zone II:** repair both slips if possible; **repair only FDP (excise one slip of FDS)** when the canal is too tight.
- **Rehabilitation:**

  - **Early active motion** (**Saint John protocol, Manchester short splint**, true active flexion with 4+ strand repairs): **best range of motion**; slightly higher rupture than passive regimens in some studies.
  - **Kleinert (active extension, passive flexion with rubber bands)**, **Duran (controlled passive motion)**.
  - **Dorsal blocking splint**: **wrist neutral to slight flexion (0-20 degrees), MCPJ flexed about 50-70 degrees, IPJs extended**, for about **4-6 weeks**.
  - Strengthening after about 8-12 weeks.
- **Complications: adhesions** (commonest; tenolysis after **3-6 months** once soft tissues settle and passive ROM is full), **rupture (about 4-10%, usually days 7-14; re-repair if early)**, **joint contracture (PIPJ flexion)**, **bowstringing**, **trigger finger**, **quadriga** (over-advanced FDP repair restricts the other fingers' flexion), **lumbrical-plus finger** (paradoxical PIPJ extension on attempted flexion when FDP is cut distal to the lumbrical origin or the graft is too long: lumbrical pulls the extensor apparatus).

4. Jersey finger (FDP avulsion; zone I)

- **Ring finger** commonest (about 75%): forced extension of an actively flexing finger (grabbing a jersey).
- **Leddy and Packer classification:**
  - **Type I:** tendon retracts into the **palm** (both vincula ruptured): **repair within 7-10 days** (poor vascularity, rapid myostatic contracture).
  - **Type II:** retracts to the **PIPJ** (long vinculum intact): repair within about **6 weeks** (commonest type).
  - **Type III: large bony fragment caught at the A4 pulley** (retracts to DIPJ): **ORIF**.

  - **Type IV:** bony fragment + tendon avulsed from the fragment: fix both.
  - **(Type V: with distal phalanx fracture.)**
- **Repair:** pull-out button, **suture anchors**.
- **Late presentation:** DIPJ arthrodesis/tenodesis, accept the deficit (FDS still flexes the PIPJ), **two-stage grafting** (rarely).

5. Flexor tendon reconstruction

- **Single-stage free tendon graft** (palmaris longus: absent in about 15%; plantaris; toe extensors) if the bed is good and pulleys intact.
- **Two-stage (Hunter) reconstruction:** stage 1: **silicone rod** inserted to form a **pseudosheath** (and pulley reconstruction), **stage 2 (about 3 months)**: tendon graft through the new sheath. For **scarred beds, damaged pulleys**.
- **Boyes preoperative classification:** grade 1 good (minimal scar) to grade 5 salvage (multiple digits, poor prognosis).
- **Pulley reconstruction:** tendon graft looped around the phalanx (**triple loop for A2**), **under the extensor mechanism for A2, over it for A4** (around the bone).

6. Extensor tendon anatomy

- **Six dorsal wrist compartments (radial to ulnar):**
  - **APL + EPB** (**De Quervain's tenosynovitis**; **separate EPB subsheath** in many: release it too),
  - **ECRL + ECRB** (**intersection syndrome**: proximal, where compartment 1 muscles cross compartment 2: about **4-8 cm proximal to Lister's tubercle**),
  - **EPL** (ulnar to **Lister's tubercle**: **attrition rupture after undisplaced distal radius fractures**, typically 4-8 weeks later),
  - **EDC + EIP** (**posterior interosseous nerve** on the floor),
  - **EDM** (overlies the DRUJ),

    - **ECU** (subsheath; ECU subluxation).
- **Juncturae tendinum** (between EDC tendons, dorsum of the hand): mask proximal EDC lacerations (finger can still extend).
- **EIP and EDM lie ulnar to the EDC tendons** of their digit (independent extension); **EIP used for EPL reconstruction** (tendon transfer).
- **Extensor mechanism of the finger:**
    - **Central slip** (EDC) inserts on the **base of the middle phalanx** (extends the PIPJ),
    - **Lateral bands** (from **interossei and lumbricals** plus EDC): join to form the **terminal tendon** at the distal phalanx (extends DIPJ),
    - **Sagittal bands** (centralise the EDC over the MCPJ; extend the MCPJ via a sling on the proximal phalanx),
    - **Triangular ligament** (dorsally, between the lateral bands: prevents volar subluxation),
    - **Transverse retinacular ligament** (prevents dorsal subluxation of lateral bands),
    - **Oblique retinacular ligament (Landsmeer):** volar to the PIPJ axis, dorsal to DIPJ: **links PIPJ and DIPJ extension** (tight in boutonniere deformity).
- **Intrinsics: flex the MCPJ and extend the IPJs** (lumbricals originate from FDP: **the only muscles that relax their own antagonist**).

7. Extensor tendon zones (Verdan)

Zone	Location	Classic injury
I	**DIPJ**	**Mallet finger**
II	Middle phalanx	Lateral band / terminal tendon lacerations
III	**PIPJ**	**Boutonniere (central slip)**
IV	Proximal phalanx	Partial lacerations (broad tendon)

Zone	Location	Classic injury
V	**MCPJ**	**Fight bite (human bite)**; **sagittal band rupture** ("**boxer's knuckle**")
VI	Metacarpal	Lacerations proximal to the juncturae
VII	**Wrist (extensor retinaculum)**	Retinacular adhesions
VIII	Distal forearm	Musculotendinous junction
IX	Proximal forearm (muscle bellies)	

- **Odd zones over joints** (I, III, V, VII); thumb T I (IPJ) to T V.

8. Mallet finger (zone I)

- **Forced flexion of an extended DIPJ**: terminal tendon rupture or bony avulsion of the dorsal base of the distal phalanx. **Untreated: swan-neck deformity** (extensor imbalance: PIPJ hyperextension).
- **Doyle classification:**
  - **Type I:** closed, ± small avulsion fracture (commonest),
  - **Type II:** laceration at/proximal to the DIPJ,
  - **Type III:** deep abrasion with loss of skin and tendon,
  - **Type IV:** mallet fracture: **A** transepiphyseal (children: **Seymour fracture** if open with nail bed injury: **requires irrigation, debridement, nail-bed repair, antibiotics**: risk of osteomyelitis); **B** hyperflexion injury with fracture **20-50% of the articular surface**; **C** hyperextension with **over 50% articular surface** and **volar subluxation of the distal phalanx**.
- **Treatment:**
  - **Most closed mallet injuries (tendinous or bony): continuous extension splinting of the DIPJ (slight hyperextension) for 6-8 weeks full-time**, then night splint for 2-4 weeks (**PIPJ left free**);

any flexion during this time restarts the clock. Effective even if presentation is delayed (up to about 3 months).

- **Surgery (relative): volar subluxation of the distal phalanx** and/or **large fragment (over about 30-50% of the articular surface)**: **extension block pinning (Ishiguro)**, ORIF. Open injuries: repair.
- **Chronic with swan neck: central slip tenotomy (Fowler)**, **spiral oblique retinacular ligament reconstruction (SORL)**, DIPJ arthrodesis.

9. Boutonniere deformity (zone III)

- **Central slip rupture**: PIPJ flexion; lateral bands migrate **volar to the PIPJ axis** (triangular ligament attenuates): become **PIPJ flexors** and **hyperextend the DIPJ**.
- **Acute injury: Elson test** (**most reliable**: PIPJ flexed 90 degrees over the table edge; attempt to extend the middle phalanx against resistance: **with central slip rupture, the DIPJ goes rigid/extends** (lateral bands act); with an intact slip, the DIPJ remains floppy).
- **Treatment:**
  - **Closed acute: PIPJ extension splinting for 6 weeks, DIPJ free (active DIPJ flexion exercises to pull the lateral bands dorsally)**.
  - **Open laceration/large avulsion fracture:** repair/fix.
  - **Chronic: supple: splinting/serial casting first** (most improve); surgical reconstruction (**Fowler/Dolphin terminal tendon tenotomy** for DIPJ hyperextension; central slip reconstruction) only once **passive PIPJ extension is restored**.
    Pseudoboutonniere = PIPJ hyperextension injury with volar plate scarring (flexion contracture without DIPJ hyperextension).

10. Sagittal band rupture ("boxer's knuckle", zone V)

- Usually **radial sagittal band** (middle finger commonest): **ulnar subluxation of EDC** with MCPJ flexion; extension lag at the MCPJ.
- **Acute: sagittal band (relative motion/yoke) splint** for about 6 weeks; **chronic/failed:** surgical repair or **centralisation (sagittal band reconstruction)**.

11. Fight bite (clenched-fist injury, zone V)

- Small wound over the MCPJ: **assume joint penetration and tendon injury**; skin, tendon and capsule wounds move apart when the fingers are extended.
- **Organisms: mixed: *Streptococcus*, *S. aureus*, *Eikenella corrodens* (classic), anaerobes**: **co-amoxiclav (amoxicillin-clavulanate)**; *Eikenella* is resistant to first-generation cephalosporins/clindamycin.
- **Exploration, irrigation and debridement**, leave wound open, splint; tendon repair delayed if contaminated.

12. Other extensor problems

- **EPL rupture after distal radius fracture** (often undisplaced): **EIP to EPL transfer** (commonest) or free graft.
- **Vaughan-Jackson syndrome** (rheumatoid arthritis): **attrition rupture of the ulnar extensor tendons (EDM, then EDC to little and ring) over a dorsally prominent ulnar head (caput ulnae)**: **tenosynovectomy, Darrach/distal ulna procedure, tendon transfers (EIP or side-to-side)**. **Mannerfelt-Norman lesion**: **FPL attrition rupture over a scaphoid osteophyte** (rheumatoid).
- **Extensor repair: zones IV-VIII**: core sutures (thinner tendons in zones I-IV need running/horizontal mattress or figure-of-eight sutures); **relative motion extension splinting (yoke)** for zones IV-VII allows early active motion; **static or dynamic splinting** otherwise.

Exam pearl: **Zone II (A1 pulley to FDS insertion) = no man's land. A2 and A4 pulleys prevent bowstringing; thumb oblique pulley most important. Repair strength depends on the number of core strands; 4-6 strand + epitendinous allows early active motion. Weakest days 5-21; ruptures usually days 7-14. Adhesions = commonest complication.**

Exam pearl: **Jersey finger: ring finger; Leddy-Packer I (palm, both vincula) repair within 7-10 days; III bony fragment caught at A4. Lumbrical-plus finger: paradoxical PIPJ extension. Quadriga: over-advanced FDP.**

Exam pearl: **Mallet: 6-8 weeks continuous DIPJ extension splint (PIPJ free); fix if volar subluxation. Boutonniere: Elson test; PIPJ splinted in extension, DIPJ free to flex. Fight bite: Eikenella, co-amoxiclav, explore. EPL rupture after undisplaced distal radius fracture: EIP transfer. Vaughan-Jackson = ulnar extensors over the ulnar head (RA); Mannerfelt = FPL over the scaphoid.**

Written by Claude for Ortho preSSaleel · updated 2026-11-14. Check doubtful points in a standard textbook.